ECON602 TA Notes: Continuous-time Methods

Haoran Wang

January 27, 2025

Contents

This lecture note introduces the continuous-time dynamic programming and its applications in macroeconomics. In a few weeks, some lectures will be based on continuous time and this lecture note helps you prepare for them.

Continuous-time methods are generally more widely applied in financial modeling (e.g. Black-Scholes), because financial data are much more high-frequency than macroeconomic data, and hence it is natural to frame high-frequency financial decision making process in continuous time, which can be taken easily to high-frequency data. In macroeconomics, however, the motivations for introducing continuous-time methods are mainly out of theoretical and computational consideration:

- Compared to discrete-time models, continuous-time models are surprisingly more inductive to analytical solutions. Take a look at the works of Fernando Alvarez and Francesco Lippi on sufficient statistics of price adjustment moments and effects of monetary policy: analytical solutions are delivered with a continuous-time formulation of dynamic price adjustment problem we saw in Boragan's part.
- The recent HANK or HACT (Heterogeneous-agent Continuous-time) literature suggests that continuous-time methods have large computational benefit. The idea is to express the economy with a system of partial differential equations (PDE), and then apply the algorithms from numerical PDE (e.g. finite difference method) to solve the dynamics. If you are interested in the literature, go to Ben Moll, Gianluca Violante or Jesus Fernandez-Villavarde's website for more details.

1 From Discrete Time to Continuous Time

Recurring Example: Discrete-time Consumption-saving Problem

Consider a standard consumption-saving problem, written in discrete time:

$$\max_{\{c_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t) \tag{1}$$

subject to

$$c_t + a_{t+1} \leq y_t + (1+r)a_t \tag{2}$$

and borrowing constraint

$$a_{t+1} \geq 0 \tag{3}$$

and a_0 is given. For simplicity, I am muting the uncertainty for now, so that the income path $\{y_t\}_{t=0}^{\infty}$ is deterministic.

Recall that we have two ways of solving this problem. The first is to use the Lagrangian method:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left[u(c_t) - \lambda_t(c_t + a_{t+1} - y_t - (1+r)a_t)\right] \tag{4}$$

and take FOCs on c_t, a_{t+1} and λ_t.

The other way is to use the recursive formulation by writing down the Bellman equation

$$V(a) = \max_{c,a'} u(c) + \beta V(a') \tag{5}$$
$$\text{s.t. } c + a' \leq y + (1+r)a \text{ and } a' \geq 0$$

which results in a value function $V(a)$ and policy functions $c(a)$ and $a'(a)$.

Continuous-time Analogue

As you will see shortly, there are continuous-time analogues for Lagrangian (called Hamiltonion) and Bellman equation (Hamilton-Jacobi-Bellman equation, or HJB) for the above two discrete-time method that we just wrote down, but before that, let me briefly discuss how one could transform a discrete-time problem (1), (2) and (3) into a continuous-time problem.

The continuous-time analogue of the above problem looks as follows. The consumer chooses function $c(t)$ and $a(t)$ to solve

$$\max_{\{c(t),a(t)\}} \int_0^\infty \exp(-\rho t) u(c(t))\, dt \tag{6}$$

subject to

$$\dot{a}(t) = ra(t) + y(t) - c(t) \tag{7}$$

and

$$a(t) \geq 0 \tag{8}$$

with $a(0)$ given.

Let's look into the details:

- The objective function (6): we replace the summation over time in (1) by integration over time. In order for the integration to be well defined, we need to place some more technical restrictions for our solution $c(t)$ (e.g. being continuously differentiable - I am not going to discuss it in detail). The instantaneous discount rate in (6) is ρ [Note: it is not discount factor, which refers to β in (1)]. Since we know that

$$\exp(-\rho t) = \frac{1}{\exp(\rho t)} \approx \frac{1}{(1+\rho)^t}$$

which is essentially the β^t in the discrete-time formulation (1). In other words, $\beta = \frac{1}{1+\rho}$.

- The budget constraint (7): here, $\dot{a}(t)$ is a new notation, largely following from the ODE/PDE literature, for

$$\dot{a}(t) \equiv \frac{da(t)}{dt} \equiv \lim_{\Delta t \to 0} \frac{a(t+\Delta t) - a(t)}{\Delta t}$$

It is important to know how to transform the discrete-time budget constraint (2) to (7). The way I view it is to interpret (2) as a special case of discrete-time budget constraint with period length Δt:

$$c_t \Delta t + a_{t+\Delta t} \leq (1 + r\Delta t)a_t + y_t \Delta t \tag{9}$$

where

  - $c_t \Delta t$ is the consumption flow from time t to time $t + \Delta t$; similar for $y_t \Delta t$.
  - With a little abuse of notation, the interest rate r is the instantaneous real interest rate, and hence $r\Delta t$ is the real interest rate between time t and time $t + \Delta t$.
  - Read (9) as "time $t + \Delta t$ asset holding $a_{t+\Delta t}$ is equal to the sum of time-t asset holding a_t plus interest income $r\Delta a_t$ and the saving from t to $t + \Delta t$, i.e. $y_t \Delta t - c_t \Delta t$".
  - In the case of (2), the period length $\Delta t = 1$.

and then rearrange terms of (9):

$$\frac{a_{t+\Delta t} - a_t}{\Delta t} \leq r a_t + y_t - c_t \tag{10}$$

After taking limit $\Delta t \to 0$ on both sides of (10), we get the continuous-time budget constraint (7).

2 Continuous-time Dynamic Programming: Deterministic

2.1 General Setup

Now I introduce the general setup of continuous-time dynamic programming. Any deterministic optimal control problems in continuous time that starts with the initial value

$x(0) = x_0$ can be expressed as

$$V(x_0) = \max_{z(t)} \int_0^\infty \exp(-\rho t) h(x(t), z(t))\, dt \tag{11}$$

subject to a law of motion for the state

$$\dot{x}(t) = g(x(t), z(t)) \tag{12}$$

where ρ is the subjective discount rate, $x \in X \subset \mathbb{R}^m$ is the state vector, and $z \in Z \subset \mathbb{R}^n$ is the control vector.

2.2 Method 1: Hamiltonian

As I mentioned, Hamiltonian is basically the continuous-time analogue of Lagrangian in the discrete-time model. Define Hamiltonian as

$$\mathcal{H}(x, z, \lambda) = h(x, z) + \lambda \cdot g(x, z) \tag{13}$$

where $\lambda \in \mathbb{R}^m$ is akin to the Lagrange multiplier in discrete time. Sometimes in the literature, λ is called a "costate". $\lambda \cdot g(x, z)$ is the inner product of vector λ and vector $g(x, z)$, both of which are m-dimensional.

It turns out that the necessary and sufficient conditions for the optimum can be expressed neatly with Hamiltonian (13), which are listed below:

1. $\mathcal{H}_z(x, z, \lambda) = 0$;
2. the law of motion for the costate λ satisfy

$$\dot{\lambda}(t) = \rho\lambda(t) - \mathcal{H}_x(x(t), z(t), \lambda(t)) \tag{14}$$

3. the law of motion for the state x satisfy

$$\dot{x}(t) = g(x(t), z(t)) \tag{15}$$

4. Initial value and boundary value: the boundary value is given by the transversality condition

$$\lim_{T \to \infty} \exp(-\rho T) \lambda(T) \cdot x(T) = 0 \tag{16}$$

and initial value is given by $x(0) = x_0$.

Simplified Proof by Lagrangian

Similar to the discrete-time problem, the constrained optimization problem in continuous time can also be solved by Lagrange multiplier:

$$\mathcal{L} = \int_0^\infty \exp(-\rho t)\left[h(x(t), z(t)) + \lambda(t) \cdot (g(x(t), z(t)) - \dot{x}(t))\right] dt \tag{17}$$

Similar to the discrete-time Lagrangian, the next step would be to take FOCs with respect to $x(t)$, $z(t)$ and $\lambda(t)$. The latter two FOCs are relatively easier:

$$h_z(x(t), z(t)) + \lambda(t) g_z(x(t), z(t)) = 0 \tag{18}$$

and

$$g(x(t), z(t)) - \dot{x}(t) = 0 \tag{19}$$

The FOC on $x(t)$ can be a bit tricky, since we don't know how to deal with $\dot{x}(t)$. Here we need to use the trick of integration by parts: the last term of (17) can be written as

$$\begin{aligned}
\int_0^\infty - \exp(-\rho t)\lambda(t)\dot{x}(t)\, dt &= \int_0^\infty - \exp(-\rho t)\lambda(t)\frac{dx(t)}{dt}\, dt \\
&= \int_0^\infty - \exp(-\rho t)\lambda(t) dx(t) \\
&= - \exp(-\rho t)\lambda(t) \cdot x(t)\Big|_0^\infty + \int_0^\infty x(t)\, d\exp(-\rho t)\lambda(t) \\
&= - \exp(-\rho t)\lambda(t) \cdot x(t)\Big|_0^\infty + \int_0^\infty x(t) \cdot \Big(\exp(-\rho t)\dot{\lambda}(t) - \rho\exp(-\rho t)\lambda(t)\Big)\, dt
\end{aligned} \tag{20}$$

From (20), we know that

- The first term,

$$- \exp(-\rho t)\lambda(t) \cdot x(t)\Big|_0^\infty = - \lim_{t\to\infty} \exp(-\rho t)\lambda(t) \cdot x(t) + \lambda(0) \cdot x(0)$$

is a constant (if the transversality condition is satisfied, so that the limit exists);

- The second term can be combined with other terms of (17), and hence we get

$$\mathcal{L} = \int_0^\infty \exp(-\rho t)\left[h(x(t), z(t)) + \lambda(t) \cdot g(x(t), z(t)) + x(t) \cdot \left(\dot{\lambda}(t) - \rho\lambda(t)\right)\right] dt \quad (21)$$
$$+ \text{constant}$$

Hence, take FOC on $x(t)$ using the modified Lagrangian (21), we get

$$\lambda(t) \cdot g_x(x(t), z(t)) + \dot{\lambda}(t) - \rho\lambda(t) = 0 \quad (22)$$

After rearranging terms, we can see that it is exactly the same as (14).

Note:

- As you may have noticed, the steps for solving the continuous-time problem using Lagrangian method are largely the same as their discrete-time counterpart, except for the tricks of deriving the costate law of motion.
- The necessary and sufficient conditions also looks pretty much the same. For example, in a discrete-time problem, the costate law of motion (14) is basically the FOC of x_{t+1} (in the consumption-saving problem, it is the Euler equation).
- I guess the only major difference is that previously in the discrete-time optimization we take FOC on x_{t+1} while here in the continuous-time problem we take FOC on $x(t)$. This difference illustrates the difference between continuous time and discrete time: in continuous time, everything happens instantaneously, so that there is not such a clear notion of today vs tomorrow as in the discrete-time case. In other words, "today is tomorrow" in continuous time.

Caveat: if you read Chapter 7 of the Acemoglu textbook carefully, you will find that there are in fact a lot of technical issues with the methods above, and some of which can be quite involved. The presentation above is very simplistic and not mathematically rigorous. For instance, it does not touch the uniqueness and existence of the solution, and it does not show the sufficiency (I believe assuming concavity of $h(x, z)$ should be enough to guarantee sufficiency).

Example: Consumption-saving Problem

Let's walk through the continuous-time consumption-saving problem, defined by (6), (7) and (8), using Hamiltonion. First, let's look into the primitives of the problem:

- Control is $c(t)$, state is $a(t)$, both of which are one-dimensional (i.e. $m = n = 1$);
- Objective function $h(x, z)$ is now $u(c(t))$, which is independent of state $x(t)$.
- Budget constraint $\dot{x}(t) = g(x(t), z(t))$ takes a linear form, i.e.

$$\dot{a}(t) = g(a(t), c(t)) = ra(t) + y(t) - c(t)$$

Define the Hamiltonion as

$$\mathcal{H}(a, c, \lambda) = u(c) + \lambda(ra + y - c) \tag{23}$$

The solution to the continuous-time problem is characterized by

1. $\mathcal{H}_c(a(t), c(t), \lambda(t)) = 0$, i.e.
$$u'(c(t)) - \lambda(t) = 0 \tag{24}$$

2. Costate transition equation $\dot{\lambda}(t) = \rho\lambda(t) - \mathcal{H}_a(a(t), c(t), \lambda(t))$: since
$$\mathcal{H}_a(a(t), c(t), \lambda(t)) = \lambda(t)r$$
we get
$$\dot{\lambda}(t) = \rho\lambda(t) - \lambda(t)r = (\rho - r)\lambda(t) \tag{25}$$

3. State transition equation:
$$\dot{a}(t) = ra(t) + y(t) - c(t)$$

4. Transversality condition:
$$\lim_{t\to\infty} \exp(-\rho t)\lambda(t)a(t) = 0 \tag{26}$$

Combining (24) and (25) gives us the Euler equation in continuous time. (24) tells us that $\lambda(t) = u'(c(t))$, and from that we can derive $\dot{\lambda}(t)$:

$$\dot{\lambda}(t) = \frac{d\lambda(t)}{dt} = \frac{du'(c(t))}{dt} = u''(c(t))\dot{c}(t) \tag{27}$$

Plug (27) into (25):

$$u''(c(t))\dot{c}(t) = (\rho - r)u'(c(t))$$

Recall that the elasticity of intertemporal substitution (EIS) is defined as

$$\sigma(t) \equiv \frac{-u'(c(t))}{u''(c(t))}$$

and hence we get

$$\frac{\dot{c}(t)}{c(t)} = (r - \rho)\sigma(t) \tag{28}$$

Equation (28) is the continuous-time Euler equation. The following analyses are immediate from (28):

- The left hand side of (28) is the instantaneous consumption growth;
- If we mute the variations in EIS by assuming CRRA (so that $\sigma(t) = \sigma$), we can see that the right hand side of (28) is a constant. In other words, (28) plus CRRA guarantees the optimal consumption to be along the balanced growth path (you will know what this is shortly) or at the steady state.
- The dynamics of consumption hinges on $r - \rho$ (if we assume CRRA):
  - if $r - \rho > 0$, which maps to $\beta(1 + r) > 1$ in discrete time, consumption will grow at rate $(r - \rho)\sigma$ along the path.
  - if $r - \rho = 0$, which maps to $\beta(1 + r) = 1$ in discrete time, consumption will be constant over time and at its steady-state value.
  - if $r - \rho < 0$, which maps to $\beta(1 + r) < 1$ in discrete time, consumption will be shrinking over time.

2.3 Method 2: Hamilton-Jacobi-Bellman Equation

Apart from the Lagrangian method, we can also use a recursive method to tackle the continuous-time dynamic programming problem. The continuous-time version of Bellman equation associated with the general problem defined by (11) and (12) is called the Hamilton-Jacobi-Bellman Equation, or HJB:

$$\rho V(x(t)) = \max_{z(t)} h(x(t), z(t)) + \frac{dV(x(t))}{dx} \cdot \dot{x}(t) \tag{29}$$
$$\text{s.t. } \dot{x}(t) = g(x(t), z(t))$$

Note that $\frac{dV(x(t))}{dx}$ is a $m-$dimensional vector and so is $\dot{x}(t)$. The second term on the right hand side is the inner product of these two vectors.

The optimality conditions associated with HJB (29) are largely the same as those for the Bellman equation in discrete time. First, replace the law of motion for state $x(t)$ into the HJB and get

$$\rho V(x(t)) = \max_{z(t)} h(x(t), z(t)) + \frac{dV(x(t))}{dx} \cdot g(x(t), z(t))$$

We have a first-order condition for the control variable $z(t)$:

$$h_z(x(t), z(t)) + \frac{dV(x(t))}{dx} \cdot g_z(x(t), z(t)) = 0 \tag{30}$$

and an envelope condition:

$$\rho \frac{dV(x(t))}{dx} = h_x(x(t), z(t)) + \frac{dV(x(t))}{dx} \cdot g_x(x(t), z(t)) + \frac{d^2V(x(t))}{dx^2} g(x(t), z(t)) \tag{31}$$

Equation (30), (31) together with the law of motion for state (12) define the value function $V(x(t))$ and policy functions $x(t), z(t)$.

Derivation of HJB

Like the Bellman equation, HJB also stems from the Bellman Optimality Condition: if a plan $\{z(t), x(t)\}$ is optimal, then the value function $V(x(t))$ must satisfy

$$V(x(t)) = \max_{z(t)} \int_t^{t+\Delta t} \exp(-(s-t)\rho) h(x(s), z(s))\, ds + \exp(-\Delta t \rho) V(x(t+\Delta t)) \tag{32}$$

for any given Δt. The discrete-time Bellman equation can be regarded as setting $\Delta t = 1$.

To derive HJB, we first rearrage terms:

$$V(x(t)) - \exp(-\Delta t \rho) V(x(t+\Delta t)) = \max_{z(t)} \int_t^{t+\Delta t} \exp(-(s-t)\rho) h(x(s), z(s))\, ds$$

and then divide Δt on both sides:

$$\frac{V(x(t)) - \exp(-\Delta t \rho) V(x(t+\Delta t))}{\Delta t} = \max_{z(t)} \frac{1}{\Delta t} \int_t^{t+\Delta t} \exp(-(s-t)\rho) h(x(s), z(s))\, ds$$

The next step is to take limit $\Delta t \to 0$ on both sides. According to the fundamental theorem of calculus, the right hand side would be

$$\max_{z(t)} \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int_{t}^{t+\Delta t} \exp(-(s-t)\rho) h(x(s), z(s))\, ds = \max_{z(t)} h(x(t), z(t))$$

The left hand side becomes

$$\begin{aligned}
\lim_{\Delta t \to 0} \frac{V(x(t)) - \exp(-\Delta t \rho) V(x(t+\Delta t))}{\Delta t} &= \lim_{\Delta t \to 0} \frac{V(x(t)) - V(x(t+\Delta t)) + (1-\exp(-\Delta t \rho)) V(x(t+\Delta t))}{\Delta t} \\
&= - \underbrace{\frac{dV(x(t))}{dt}}_{= \frac{dV(x(t))}{dx} \cdot \dot{x}(t)} + \rho V(x(t))
\end{aligned}$$

where the second term follows from L'Hopital's rule

$$\lim_{\Delta t \to 0} \frac{1 - \exp(-\rho \Delta t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{\rho \exp(-\rho \Delta t)}{1} = \rho$$

Putting everything together, we have

$$-\frac{dV(x(t))}{dx} \cdot \dot{x}(t) + \rho V(x(t)) = \max_{z(t)} h(x(t), z(t))$$

which is the same as (29).

The Consumption-saving Example Again

We can easily write down the HJB for the consumption-saving problem:

$$\begin{aligned}
\rho V(a(t)) &= \max_{c(t)} u(c(t)) + \frac{dV(a(t))}{da} \cdot \dot{a}(t) \\
\text{s.t. } \dot{a}(t) &= ra(t) + y(t) - c(t)
\end{aligned} \tag{33}$$

FOC is

$$u'(c(t)) = \frac{dV(a(t))}{da} \tag{34}$$

Envelope condition:

$$\rho \frac{dV(a(t))}{da} = \frac{dV(a(t))}{da} r + \frac{d^2 V(a(t))}{da^2} \dot{a}(t) \tag{35}$$

We can derive the Euler equation from the FOC and the envelope condition. First, take derivatives with respect to t for (34):

$$u''(c(t))\dot{c}(t) = \frac{d^2V(a(t))}{da^2}\dot{a}(t)$$

From (35), we know that

$$\frac{d^2V(a(t))}{da^2}\dot{a}(t) = (\rho - r)\frac{dV(a(t))}{da} = (\rho - r)u'(c(t))$$

Hence, we have

$$u''(c(t))\dot{c}(t) = (\rho - r)u'(c(t))$$

or equivalently

$$\frac{\dot{c}(t)}{c(t)} = (r - \rho)\sigma(t)$$

which is the same as the Euler equation derived from Hamiltonian.

3 Continuous-time Dynamic Programming: Stochastic, Poisson Process

We now extend the deterministic benchmark to a stochastic setting. We will first look into a simple case of Poisson income process for the continuous-time consumption-saving problem above. This benchmark case studies an income process $\{y(t)\}$ that follows a continuous-time discrete Markov chain (i.e. the support of $y(t)$ is countable).

Continuous-time Markov Chain and KFE

Suppose there are in total J possible values that income $y(t)$ could take: $\{y^1, \cdots, y^J\}$. The Poisson process is defined by the instantaneous transition hazard

$$\lim_{\Delta t \to 0} P(y(t + \Delta t) = y^{j'} | y(t) = y^j) = \lambda_{jj'}$$

i.e. if at time t the state is y^j, the Poisson arrival rate of state $y^{j'}$, or, more intuitively, the probability of next instant $t + \Delta t$ (Δt is very small) state being $y^{j'}$, is equal to $\lambda_{jj'}$. We can stack these $\lambda_{jj'}$ into a Markov transition matrix

$$\begin{bmatrix} 1-\sum_{j\neq 1}\lambda_{1j} & \lambda_{12} & \cdots & \lambda_{1J} \\ \lambda_{21} & 1-\sum_{j\neq 2}\lambda_{2j} & \cdots & \lambda_{2J} \\ \vdots & \vdots & \vdots & \vdots \\ \lambda_{J1} & \lambda_{J2} & \cdots & 1-\sum_{j\neq J}\lambda_{Jj} \end{bmatrix}$$

which can be written as

$$I + \Lambda$$

where Λ is defined as

$$\Lambda \equiv \begin{bmatrix} -\sum_{j\neq 1}\lambda_{1j} & \lambda_{12} & \cdots & \lambda_{1J} \\ \lambda_{21} & -\sum_{j\neq 2}\lambda_{2j} & \cdots & \lambda_{2J} \\ \vdots & \vdots & \vdots & \vdots \\ \lambda_{J1} & \lambda_{J2} & \cdots & -\sum_{j\neq J}\lambda_{Jj} \end{bmatrix} \tag{36}$$

As you can see, this is simply the continuous-time version of a Markov chain. Many properties of the discrete-time Markov chain carry over to the continuous Markov chain. For instance, suppose the distribution of income at time t is denoted by $f(t)$. We can easily derive the distribution of income at period $t + \Delta t$:

$$f(t+\Delta t) = (I + \Lambda\Delta t)' f(t) = f(t) + \Lambda' f(t)\Delta t \tag{37}$$

Move $f(t)$ to the left hand side, divide both sides by Δt, and then take limit $\Delta t \to 0$:

$$\dot{f}(t) = \Lambda' f(t) \tag{38}$$

Note, this is a law of motion for a probability distribution. In many contexts, such equations are called "kolmogorov forward equation" (KFE).

Derivation of HJB with Poisson Process

We next derive the HJB associated with the Poisson income process. Note that now since there is an exogenous state $y(t)$, the value function has two arguments: asset position a and income y.

It turns out that the HJB takes the form

$$\rho V(a, y^j) = \max_c u(c) + V_a(a, y^j)\dot{a} + \sum_{j' \neq j} \underbrace{\lambda_{jj'}}_{\equiv P(y^{j'}|y^j)} [V(a, y^{j'}) - V(a, y^j)] \tag{39}$$

We can see that compared with the HJB under certainty, there is an additional term on the right hand side.

For simplicity, I derive the case for the two-state Poisson process case, i.e. $J = 2$. Again, start from the discrete-time formulation:

$$V(a(t), y^1) = \max_c u(c) + \beta[(1 - \lambda_{12})V(a(t+1), y^1) + \lambda_{12}V(a(t+1), y^2)] \tag{40}$$

Generalize the time interval to Δt:

$$V(a(t), y^1) = \max_c u(c)\Delta t + \exp(-\rho\Delta t)[(1 - \lambda_{12}\Delta t)V(a(t+\Delta t), y^1) + \lambda_{12}\Delta t V(a(t+\Delta t), y^2)] \tag{41}$$

Rearrange the terms, we get

$$\frac{V(a(t), y^1) - \exp(-\rho\Delta t)V(a(t+\Delta t), y^1)}{\Delta t} = \max_c u(c) + \exp(-\rho\Delta t)\lambda_{12}[V(a(t+\Delta t), y^2) - V(a(t+\Delta t), y^1)] \tag{42}$$

Lastly, take $\Delta t \to 0$ on both sides. LHS becomes

$$-V_a(a, y^1)\dot{a}(t) + \rho V(a(t), y^1)$$

and RHS becomes

$$\max_c u(c) + \lambda_{12}[V(a(t), y^2) - V(a(t), y^1)]$$

Equate them, and we get equation (40).

Food for Thought

If you have some spare time, think about the following questions:

- What about $J \to \infty$, i.e. we have a continuous distribution?
- What is the Kolmogorov Forward Equation for the joint distribution of (a, y)? This is the continuous-time version of the law of motion for wealth distribution.

I will cover the first bullet point next week.

4 Continuous-time Dynamic Programming: Stochastic, Diffusion Process

Now we look into the stochastic continuous-time dynamic programming problem with a continuous stochastic process.

4.1 Derive HJB Again

We revisit how to go from the discrete-time Bellman equation to the continuous-time Hamilton-Jacobi-Bellman equation. First, write down the discrete-time Bellman equation for period length Δt:

$$V(a(t), y(t)) = \max_{c(t)} u(c(t))\Delta t + \exp(-\rho\Delta t)\mathbb{E}_t[V(a(t+\Delta t), y(t+\Delta t))]$$

Then, minus $(1-\exp(-\rho t))V(a(t), y(t))$ on both sides:

$$V(a(t), y(t))(1-\exp(-\rho\Delta t)) = \max_{c(t)} u(c(t))\Delta t + \exp(-\rho\Delta t)\underbrace{\mathbb{E}_t[V(a(t+\Delta t), y(t+\Delta t)) - V(a(t), y(t))]}_{\equiv\mathbb{E}[dV]}$$

Lastly, divide both sides by Δt and take limit $\Delta t \to 0$:

$$\rho V(a(t), y(t)) = \max_{c(t)} u(c(t)) + \frac{\mathbb{E}_t[dV]}{dt} \tag{43}$$

Note that here, I put a subscript on the expectation operator. This is because the expectation in time t is in fact a conditional expectation that takes into account all the information in all precedent periods $s \leq t$, namely, $\{a(s)\}_{s\leq t}$ and $\{y(s)\}_{s\leq t}$. And importantly, $a(t)$ and $y(t)$ are stochastic processes, and therefore, computing the $\mathbb{E}_t[dV]$ is essentially taking integration over a stochastic process, and the value function V itself is a stochastic process as well. Equations like (43) is usually called a stochastic differential equation (SDF) as it captures how a stochastic process V evolves over time.

It turns out that the standard rules in deterministic calculus no longer holds when we are dealing with stochastic processes, and the mathematics behind it can be quite involved. The approach taken by this note is largely heuristic: I will refrain from going into the technical details (such as using languages in measure theory) as much as possible.

4.2 Brownian Motion and Stochastic Differential Equation

We start with Brownian motion. A Brownian motion (or Wiener Process) is a stochastic process $W(t)$ where

1. $W(0) = 0$;
2. If $s_1 < t_1 < s_2 < t_2 < \cdots < s_n < t_n$, then $W(t_1) - W(s_1)$, $W(t_2) - W(s_2)$, $\cdots$, $W(t_n) - W(s_n)$ are independent to each other;
3. For $s < t$, $W(t) - W(s) \sim N(0, \sigma^2(t - s))$;
4. W is a continuous function of t.

The case where $\sigma = 1$ is usually called a standard Brownian motion or a standard Wiener process.

Note:

1. Essentially, a Brownian motion is the random walk in continuous time. For example, consider a binary random variable X where

$$X = \begin{cases} = 1 & \text{with prob. } 1/2 \\ = -1 & \text{with prob. } 1/2 \end{cases}$$

   You can think of it as a drunk man who randomly choose whether to move forward or backward. Suppose we have independent draws $X_1, \cdots, X_t$, and the sum $S_t = \sum_{i=1}^{t} X_t$ gives us the location of the drunk man after t moves. This variable S_t is a binomial random variable, and it turns out that as $\Delta t \to 0$, the distribution of S_t is a Brownian motion.

2. If we define $dW(t) \equiv W(t + dt) - W(t)$ when dt is small, we have

$$\mathbb{E}[dW(t)] = 0$$

$$\mathbb{E}[(dW(t))^2] = var(dW(t)) = \sigma dt$$

   and when $\sigma = 1$, $\mathbb{E}[dW] = dt$. It turns out to be an important property that will be exploited a lot in stochastic calculus.

3. Although $W(t)$ is continuous with respect to t (i.e. it has a continuous path), because of the increment is random, it is almost surely non-differentiable with respect

to t. This is the reason why the standard calculus rule no longer holds when we are evaluating something like $\int g(t)\, dW(t)$ (we will define this object more properly later): loosely speaking, in standard calculus, we work with a smooth differential (i.e. $W(t)$ is continuously differentiable).

From now on, I will use $W(t)$ to denote the standard Brownian motion/Wiener process. A more general class of stochastic process can be written as

$$X(t) = X_0 + \int_0^t \mu(X, s)\, ds + \int_0^t \sigma(X, s)\, dW(s) \tag{44}$$

As we can see, X_0 is the initial condition or the starting point of the stochastic process (it is easy to verify that $X(0) = X_0$). The $\mu(X, s)$ term is called "drift" which captures some form of the time trend of the dynamics (and note that the drift can in fact be a function of X and time; see the example below), and the term $\sigma(X, s)$ is called the "diffusion" or "volatility" which captures the "random" part along the dynamics of $X(t)$. It is a convention to express (44) in the form of a stochastic differential equation

$$dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dW(t), \text{ with } X(0) = X_0 \tag{45}$$

Examples Let's look at several examples.

Example 1: Brownian motion with drift

First, consider the case where $\mu(X, t) = \mu$ and $\sigma(X, t) = \sigma$, i.e. the drift and the volatility are both constant. The stochastic process degenerates to

$$dX(t) = \mu dt + \sigma dW(t) \tag{46}$$

or equivalently

$$X(t) = X_0 + \int_0^t \mu ds + \int_0^t \sigma dW(s) = X_0 + \mu t + \sigma W(t) \tag{47}$$

The last equality will be explained later. This is a Brownian motion with drift. Loosely speaking, $X(t)$ has a linear trend μt (i.e. in each period, $X(t)$ will be increased by μ), with some fluctuations around the trend, whose standard deviation is equal to σ. It is easy to see that

$$\mathbb{E}[X(t)] = X_0 + \mu t$$

and

$$var(X(t)) = \sigma^2$$

Example 2: Percentage change as a Brownian motion

Consider

$$dX(t) = \mu X(t)dt + \sigma X(t)dW(t)$$

or equivalently

$$\frac{dX(t)}{X(t)} = \mu dt + \sigma dW(t) \tag{48}$$

The left hand side is the percentage change of $X(t)$ across time. Define $\tilde{X}(t) \equiv \log X(t)$. It is easy to see that

$$d\tilde{X}(t) = d\log X(t) = \frac{1}{X(t)}dX(t)$$

hence we can write (48) as

$$d\tilde{X}(t) = \mu dt + \sigma dW(t)$$

which, by Example 1, is equivalent to

$$\tilde{X}(t) = \log X_0 + \mu t + \sigma W(t)$$

and hence $X(t)$ is given by

$$X(t) = \exp(\tilde{X}(t)) = \exp(\log X_0 + \mu t + \sigma W(t)) \tag{49}$$

(49) says that in the long run, $X(t)$ grows on average at a rate μ, with some fluctuations around this trend. In other words, μ is the long-run growth trend, and σ captures the volatility of the business cycles. You will see it a lot in the growth literature.

It is easy to get

$$\mathbb{E}[X(t)] = \exp\left(\log X_0 + \mu t + \frac{1}{2}\sigma^2\right)$$

and

$$var(X(t)) = \frac{\exp(2(\log X_0 + \mu t) + \sigma^2)}{\exp(\sigma^2) - 1}$$

following the formula for log-normal distribution.

Example 3: Ornstein-Uhlenbeck process

Ornstein-Uhlenbeck process is the continuous counterpart of an AR(1) process in discrete time. The stochastic differential equation takes the form

$$dX(t) = \theta(\bar{X} - X(t))dt + \sigma dW(t) \tag{50}$$

with $X(0) = X_0$. It turns out that the mean of $X(t)$ is given by

$$\mathbb{E}[X(t)] = X_0 \exp(-\theta t) + \bar{X}(1 - \exp(-\theta t))$$

We can see that as $t \to \infty$, the path would converge to the mean $\bar{X}$. This mean-reverting property is similar to an AR(1) process in discrete time. Finance people like this.

4.3 Stochastic Integration and Ito's Lemma

So far we have been writing down the integration

$$\int_0^t g(s)\, dW(s)$$

without properly defining it. Here, both $g(s)$ and $W(s)$ are stochastic processes. In this subsection I will lay out the details of how this integration is defined and computed.

Why do we care about this?

Think about the example of random walk in discrete time. Suppose in period t I am to make a bet whether the drunk man's next move is forward or backward, based on his previous moves from period 0 up to period t. For instance, suppose at time t_k, the bet amount is chosen to be $g(t_k)$, and if in the next period t_{k+1}, the drunk man's move $X(t_{k+1}) \equiv S(t_{k+1}) - S(t_k) = 1$ the bettor gets $g(t_k)$, otherwise he loses $g(t_k)$. We are interested in the total dollar amount that the bettor wins up to time t, denoted as $Z(t)$. Intuitively, $Z(t)$ can be written as the sum of all bets made by time t:

$$Z(t) = \sum_{k=0}^{n-1} g(t_k)(S(t_{k+1}) - S(t_k)) = \sum_{k=0}^{n-1} g(t_k)\Delta S(t_{k+1})$$

The right hand side of the above equation has a same structure as and is the discrete-time counterpart of the stochastic integration $\int_0^t g(s)\, dW(s)$.

An Easy Case: Step function

Similar to Riemann integral, where the definition starts from the integrand being a step function, here for stochastic calculus, the case where $g(s)$ is a step function is also an easy one. If $g(t)$ is a step function, i.e.

$$g(s) = \begin{cases} g_0 & \text{if } 0 = t_0 \leq s < t_1 \\ g_1 & \text{if } t_1 \leq s < t_2 \\ \cdots & \\ g_{n-1} & \text{if } t_{n-1} \leq s < t_n = t \end{cases}$$

The stochastic integration $\int_0^t g(s)\, dW(s)$ is defined as

$$\int_0^t g(s)\, dW(s) = \sum_{k=0}^{n-1} g_k \cdot (W(t_{k+1}) - W(t_k))$$

In our previous examples, we face the calculation of $\int_0^t \sigma\, dW(s)$. Note that the constant function is simply a special case of a step function, and therefore we have

$$\int_0^t \sigma\, dW(s) = \sigma(W(t) - W(0)) \tag{51}$$

Definition of Stochastic Integration

For stochastic processes $g(t)$ and $W(t)$, the stochastic integral $\int_0^t g(s)dW(s)$ is defined as

$$\int_0^t g(s)dW(s) = \lim_{n\to\infty} \sum_{k=0}^{n-1} g(t_k)[W(t_{k+1}) - W(t_k)] \tag{52}$$

where $0 = t_0 < t_1 < \cdots < t_n = t$ is a partition of the interval $[0, t]$. The stochastic integration is well-defined if and only if the limit on the right hand side of (52) exists. Again, this is very similar to the way that Riemann integral is generalized from the step-function benchmark to any integrable functions: what we are doing here is essentially "approximating" $g(t)$ using a step function with a very fine partition.

From now on, I make two assumptions:

1. $W(t)$ is the standard Wiener process;

2. stochastic process $g(t)$ is measurable to $\{W(s)\}_{s\leq t}$. Roughly speaking, it is a func-

tion of the past realizations of $W(t)$, so that when taking conditional expectation $\mathbb{E}[\cdot|\{W(s)\}_{s\leq t}]$, we have

$$\mathbb{E}\left[g(t)|\{W(s)\}_{s\leq t}\right] = g(t)$$

Of course, a more mathematically rigorous way to express the conditional expectation is to use filtration/sigma-algebra but I don't want to use too much measure theoretic language here.

Ito's Lemma

Ito's lemma can be regarded as the chain rule and the fundamental theorem of calculus in stochastic calculus. Given a stochastic process $X(t)$, it provides us with tools to specify how $f(X(t))$ evolves dynamically, where f is a function with two continuous derivatives. This will be useful to our HJB above: remember that we want to know how to compute $\mathbb{E}[dV(t)]$, and $V(t)$ is a function of endogenous state $a(t)$ and exogenous state $y(t)$, both of which are stochastic processes.

The plain-vanilla version of Ito's lemma studies the integration over a standard Brownian motion. If f is a function with two continuous derivatives, and $W(t)$ is a standard Brownian motion, then

$$f(W(t)) - f(W(0)) = \int_0^t f'(W(s))\, dW(s) + \frac{1}{2}\int_0^t f''(W(s))\, ds \tag{53}$$

Or, written in differential form,

$$df(W(t)) = f'(W(t))dW(t) + \frac{1}{2}f''(W(t))dt \tag{54}$$

At the first glance, it may be difficult to understand where the second-order term comes from. Here I provide a sketch of proof. First, we know that for any partition $0 = t_0 < t_1 < \cdots < t_n = t$, we have

$$f(W(t)) - f(W(0)) = \sum_{k=0}^{n-1} f(W(t_{k+1})) - f(W(t_k))$$

From Taylor expansion we know that

$$f(W(t_{k+1})) - f(W(t_k)) = f'(W(t_k))(W(t_{k+1}) - W(t_k)) + \frac{1}{2}f''(W(t_k))(W(t_{k+1}) - W(t_k))^2 + \text{h.o.t}$$

Hence

$$f(W(t))-f(W(0)) = \underbrace{\sum_{k=0}^{n-1} f'(W(t_k))(W(t_{k+1}) - W(t_k))}_{\to\int_0^t f'(W(s))\,dW(s) \text{ as } n\to\infty} + \underbrace{\frac{1}{2}\sum_{k=0}^{n-1} f''(W(t_k))(W(t_{k+1}) - W(t_k))^2}_{?} + \text{h.o.t} \tag{55}$$

The first term on the right hand side becomes $\int_0^t f'(W(s))\,dW(s)$ as the partition gets finer, i.e. $n \to \infty$, by definition (52). It remains for us to know what the second term is. Recall that for a standard Brownian motion,

$$W(t_{k+1}) - W(t_k) \sim N(0, t_{k+1} - t_k)$$

which means that

$$\mathbb{E}[(W(t_{k+1}) - W(t_k))^2] = t_{k+1} - t_k$$

and

$$\begin{aligned} var[(W(t_{k+1}) - W(t_k))^2] &= \mathbb{E}[(W(t_{k+1}) - W(t_k))^4] - \mathbb{E}[(W(t_{k+1}) - W(t_k))^2]^2 \\ &= 3(t_{k+1} - t_k)^2 - (t_{k+1} - t_k)^2 = 2(t_{k+1} - t_k)^2 \end{aligned}$$

Since $W(t_1) - W(t_0)$, $W(t_2) - W(t_1)$, $\cdots$, $W(t_n) - W(t_{n-1})$ are independent to each other, we immediately have

$$\sum_{k=0}^{n-1} \mathbb{E}[(W(t_{k+1}) - W(t_k))^2] = \sum_{k=0}^{n-1}(t_{k+1} - t_k) = t \to t \text{ as } n \to \infty$$

and

$$\begin{aligned} var\left[\sum_{k=0}^{n-1}(W(t_{k+1}) - W(t_k))^2\right] &= 2\sum_{k=0}^{n-1}(t_{k+1} - t_k)^2 \\ &\leq 2\max_k |t_{k+1} - t_k| \cdot \sum_k (t_{k+1} - t_k) \\ &= 2t\max_k |t_{k+1} - t_k| \to 0 \text{ as } n \to \infty \end{aligned}$$

Hence, we know that on the given interval $[0, t]$, we have

$$\sum_{k=0}^{n-1}(W(t_{k+1}) - W(t_k))^2 \to t \text{ a.s. as } n \to \infty \tag{56}$$

(56) turns out to be a very important result. As the partition of $[0, t]$ gets finer and finer, we find that the quadratic sum converges almost surely to the length of interval t. In fact, the same arguments could be applied to any arbitrary interval $[a, b]$.

Using this observation, we take a second look at the question mark term of (55). Note that we could partition the interval $[0, t]$ by using m grid points $\{t_0, \cdots, t_{m-1}\}$ for the function $f''(W(s))$, and then refine this partition by partitioning each sub-interval $[t_i, t_{i+1}]$ with l grid points to incorporate $dW(t)^2$. The question mark term can be written as

$$\begin{aligned}
\frac{1}{2}\sum_{k=0}^{n-1} f''(W(t_k))(W(t_{k+1}) - W(t_k))^2 &= \frac{1}{2}\sum_{i=0}^{m} f''(W(t_{il})) \underbrace{\sum_{j=0}^{l-1}(W(t_{il+j+1}) - W(t_{il+j}))^2}_{\to t_{(i+1)l} - t_{il} \text{ as } n\to\infty} \\
&= \frac{1}{2}\sum_{i=0}^{m} f''(W(t_{il}))(t_{(i+1)l} - t_{il}) \\
&\to \frac{1}{2}\int_0^t f''(W(s))\, ds
\end{aligned} \tag{57}$$

The second line follows from applying (56) to each sub-interval $[t_i, t_{i+1}]$ where $i = 0, \cdots, m-1$, and the last line follows from the definition of Riemann integral. Combining (55) and (57), we prove the basic Ito's lemma (53).

A Useful Rule of Thumb

Equation (56) and (57) give us important insights into stochastic integration. The second-order term (57) captures the first-order effect of volatility of $W(t)$ on the new stochastic process $f(W(t))$. More precisely, it comes from $(dW(t))^2$, and (56) and (57) show that it is of order dt. Hence, for the convenience of future calculation, we can use the following rule of thumb:

$$dt \cdot dt = 0, dt \cdot dW = 0, (dW)^2 = dt$$

Let's take a look at several examples.

Examples

Our first example is to derive a more general version of Ito's lemma. Suppose now we have a stochastic process $X(t)$ defined by the following stochastic differential equation

$$dX(t) = \mu(X, t)dt + \sigma(X, t)dW(t)$$

where $W(t)$ is again the standard Brownian motion. The function f is again a twice differentiable function. What is the stochastic differential equation for stochastic process $f(X(t))$?

Again, we apply the second-order approximation on f and reach a similar equation to (55):

$$f(X(t)) - f(X(0)) = \int_0^t f'(X(s))\, dX(s) + \frac{1}{2}\int_0^t f''(X(s))\, (dX(s))^2$$

Apply the above rule of thumb to get $(dX(t))^2$:

$$\begin{aligned}
(dX(t))^2 &= (\mu(X,t)dt + \sigma(X,t)dW(t))^2 \\
&= \mu(X,t)^2 dt^2 + \sigma(X,t)^2 (dW(t))^2 + 2\mu(X,t)\sigma(X,t)dt \cdot dW(t) \\
&= \sigma(X,t)^2 (dW(t))^2 = \sigma(X,t)^2 dt
\end{aligned}$$

And therefore we get

$$\begin{aligned}
f(X(t)) - f(X(0)) &= \int_0^t f'(X(s))\, (\mu(X,s)ds + \sigma(X,s)dW(s)) + \frac{1}{2}\int_0^t f''(X(s))\sigma(X,s)^2\, ds \\
&= \int_0^t \left(f'(X(s))\mu(X,s) + \frac{1}{2} f''(X(s))\sigma(X,s)^2\right) ds + \int_0^t f'(X(s))\sigma(X,s)\, dW(s)
\end{aligned}$$

or equivalently

$$df(X(t)) = \left(f'(X(t))\mu(X,t) + \frac{1}{2} f''(X(t))\sigma(X,t)^2\right) dt + f'(X(t))\sigma(X,t)dW(t) \tag{58}$$

This more general version of Ito's lemma is more commonly seen in the literature.

We can work with one example that applies (58). Suppose $f = \exp$ and

$$dX(t) = \mu dt + \sigma dW(t)$$

We directly plug exp into (58):

$$\begin{aligned}
d\exp(X(t)) &= \left(\exp(X(t))\mu + \frac{1}{2}\exp(X(t))\sigma^2\right) dt + \exp(X(t))\sigma dW(t) \\
&= \exp(X(t)) \left[\left(\mu + \frac{1}{2}\sigma^2\right) dt + \sigma dW(t)\right] \qquad (59)
\end{aligned}$$

4.4 Return to HJB

After the tedious preparation, now we are ready to work out the HJB. Apply Ito's lemma to $\mathbb{E}_t[dV]$, or, think about second-order approximation:

$$\begin{aligned}\mathbb{E}_t[dV] =& \mathbb{E}\Big[V_a(a(t),y(t))da + V_y(a(t),y(t))dy \\ &+ \frac{1}{2}V_{aa}(a(t),y(t))da^2 + \frac{1}{2}V_{yy}(a(t),y(t))dy^2 + V_{ay}(a(t),y(t))dady\Big] \end{aligned} \tag{60}$$

We can work out each term in (60):

$$\mathbb{E}_t\left[V_a da\right] = \mathbb{E}_t\left[V_a(a(t),y(t))(ra(t)+y(t)-c(t))dt\right] = V_a(a(t),y(t))(ra(t)+y(t)-c(t))dt$$

(Note, the conditional expectation can be taken off because $V_a(a(t),y(t))$ and $(ra(t)+y(t)-c(t))$ are both part of the time-t information set.)

$$\mathbb{E}\left[V_y dy\right] = \mathbb{E}\left[V_y(a(t),y(t))(\mu dt + \sigma dW_t)\right] = V_y(a(t),y(t))(\mu\mathbb{E}[dt]+\sigma\mathbb{E}[dW_t]) = V_y(a(t),y(t))\mu dt$$

(The last equality follows from the fact that $\mathbb{E}[dW(t)] = 0$ for all t, if $W(t)$ is a standard Brownian motion.)

$$\mathbb{E}\left[V_{aa}da^2\right] = V_{aa}(ra(t)-c(t)+y(t))(dt)^2 = \text{h.o.t}$$

$$\mathbb{E}\left[V_{yy}dy^2\right] = V_{yy}\mathbb{E}\left[\mu^2dt^2+\sigma^2dW_t^2+2\mu\sigma dtdW\right] = \text{h.o.t} + V_{yy}(a(t),y(t))\sigma^2dt$$

$$\mathbb{E}\left[V_{ay}dady\right] = V_{ay}(a(t),y(t))\mathbb{E}[dady] = \text{h.o.t}$$

(The above three equations follow from the rule of thumb.)

Putting everything together,

$$\begin{aligned}\mathbb{E}[dV] &= V_a(a(t),y(t))(ra(t)+y(t)-c(t))dt + V_y(a(t),y(t))\mu dt + \frac{1}{2}V_{yy}(a(t),y(t))\sigma^2dt \\ &= \left[V_a(a(t),y(t))(ra(t)+y(t)-c(t)) + V_y(a(t),y(t))\mu + \frac{1}{2}V_{yy}(a(t),y(t))\sigma^2\right]dt\end{aligned} \tag{61}$$

and hence the HJB becomes

$$\rho V(a(t),y(t)) = \max_{c(t)} u(c(t)) + V_a(a(t),y(t))(ra(t)+y(t)-c(t)) + V_y(a(t),y(t))\mu + \frac{1}{2}V_{yy}(a(t),y(t))\sigma^2 \tag{62}$$